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Chromosomes

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What are Chromosomes?

- Chromosomes are complex structures located in the cell nucleus
- Composed of DNA, histone and non-histone proteins, RNA , and polysaccharides. They are basically the "packages" that contain the DNA.
- Some examples include peripheral blood, bone marrow, amniotic fluid, and products of conception.
- Under the microscope chromosomes appear as thin, thread-like structures.
- Study of chromosomes is **Karyology**

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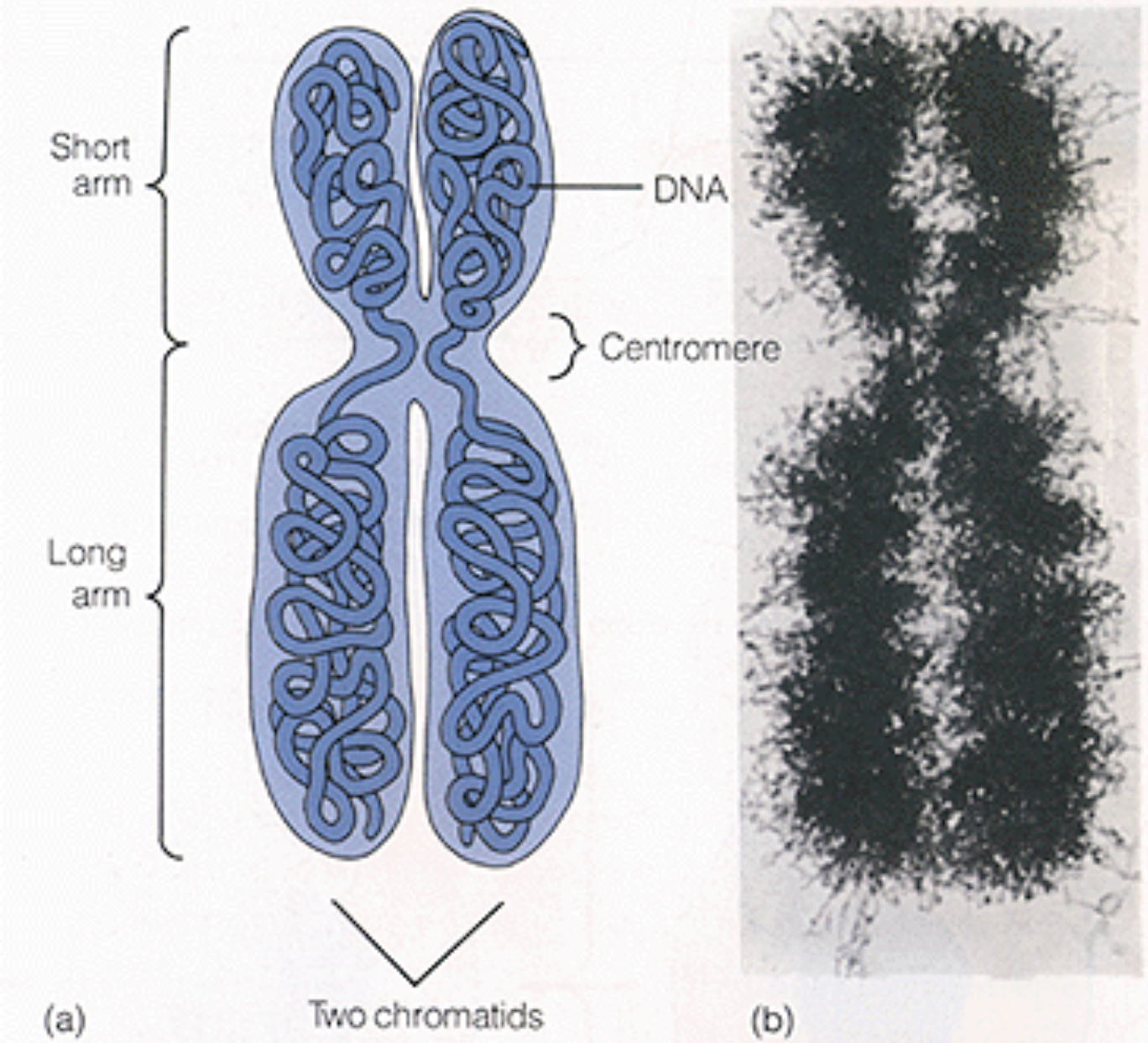
Chromosomes

- They all have a short arm and long arm separated by a primary constriction called the ***centromere.***
- The short arm is designated as ***p*** and the long arm as ***q.***
- The centromere is the location of spindle attachment and is an integral part of the chromosome.



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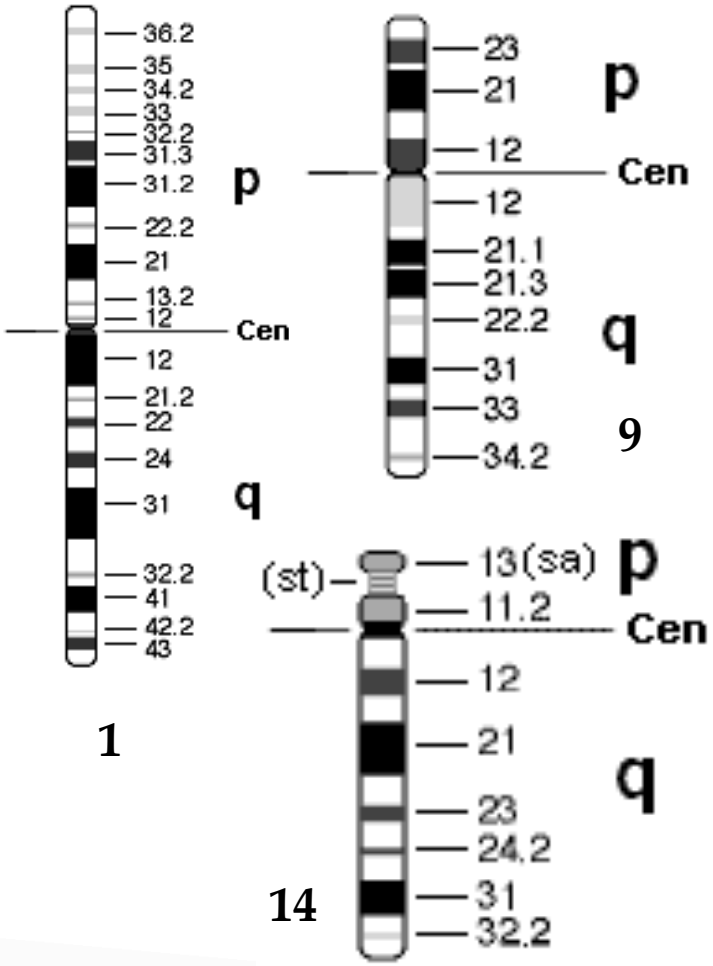
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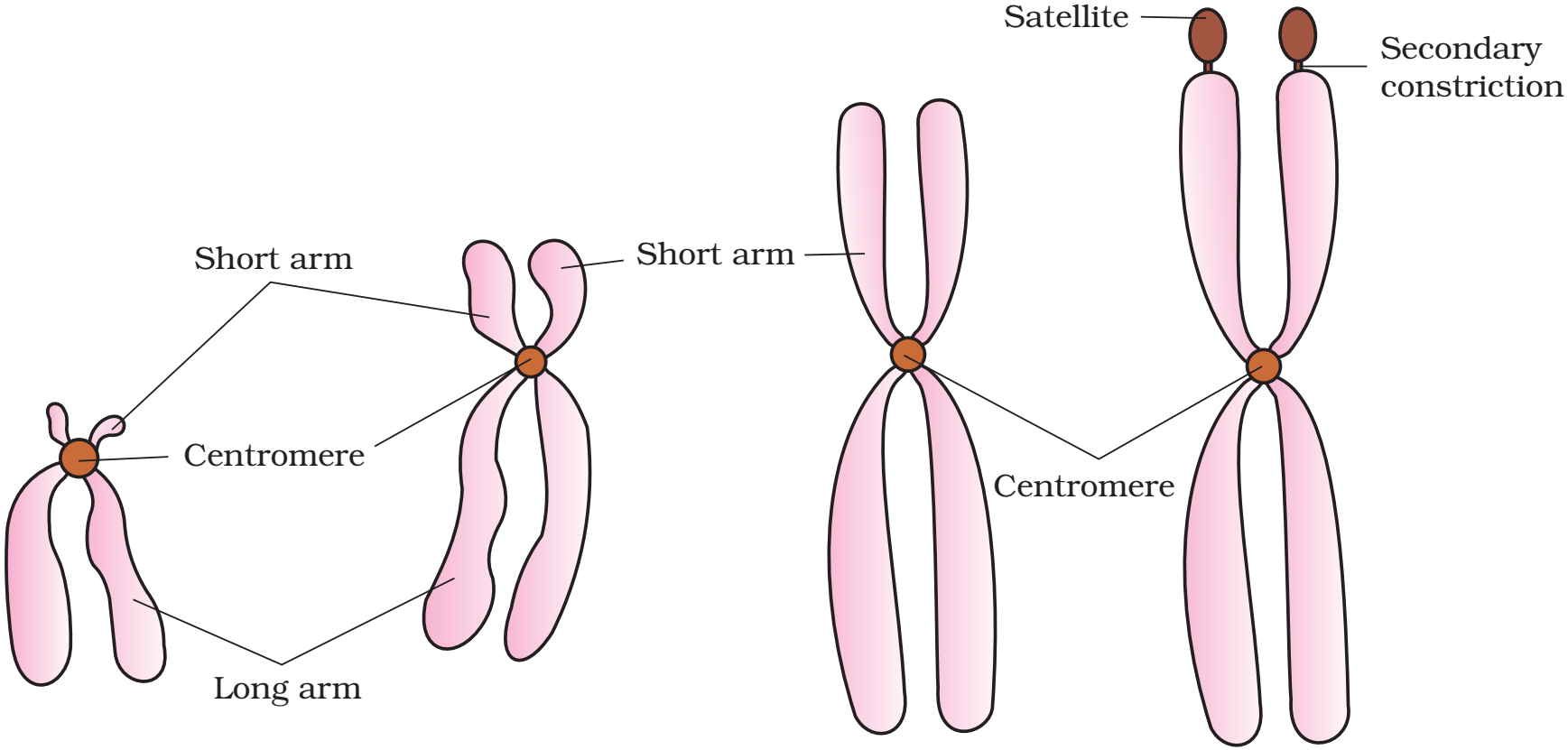
Chromosomes



- Based on the position of centromere, human chromosomes can be classified into, **Metacentric, Sub-metacentric, Acrocentric**
- Acrocentrics frequently have a secondary constrictions connecting small pieces of DNA called stalks & satellites

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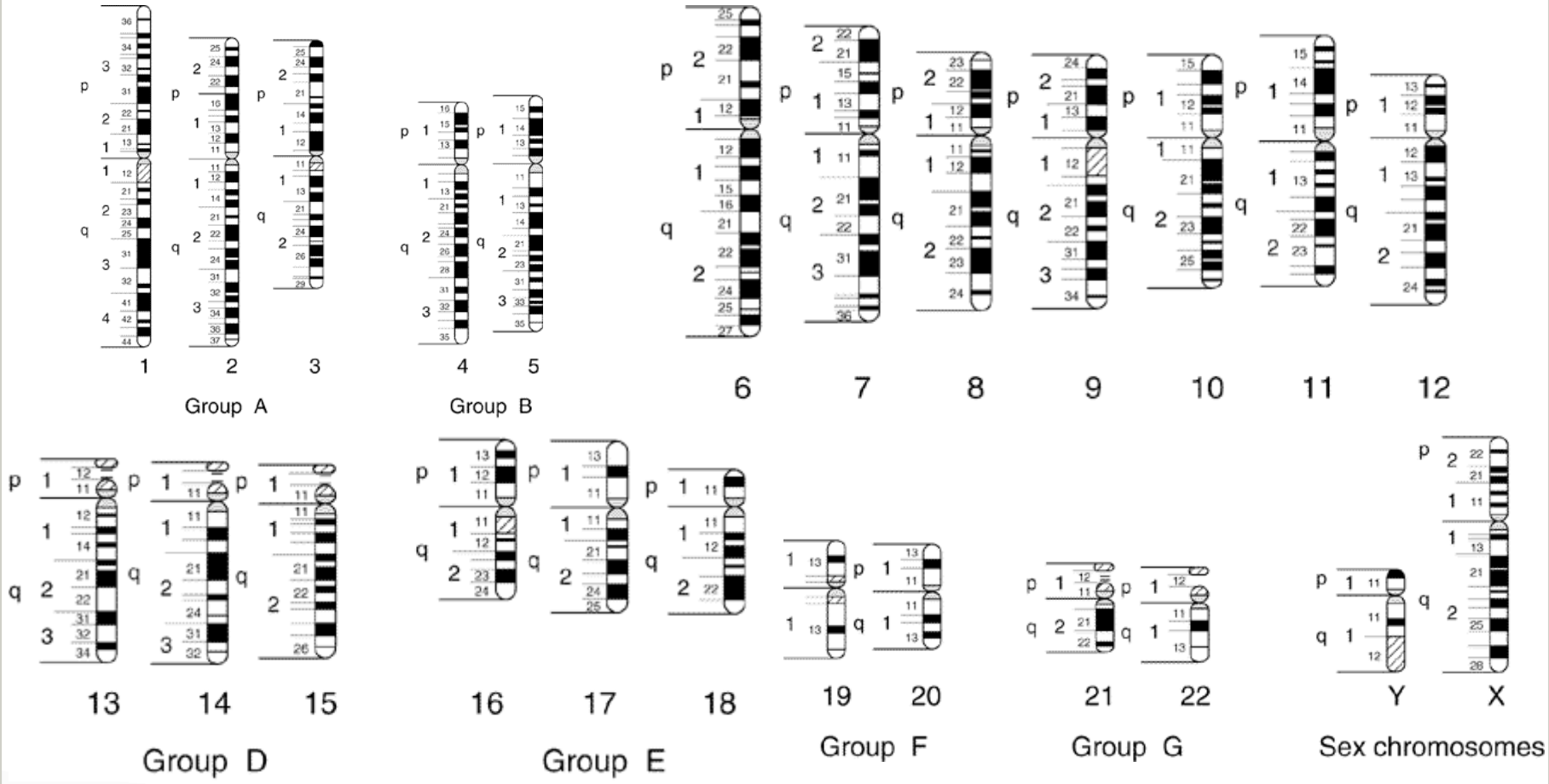
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Human Ideogram

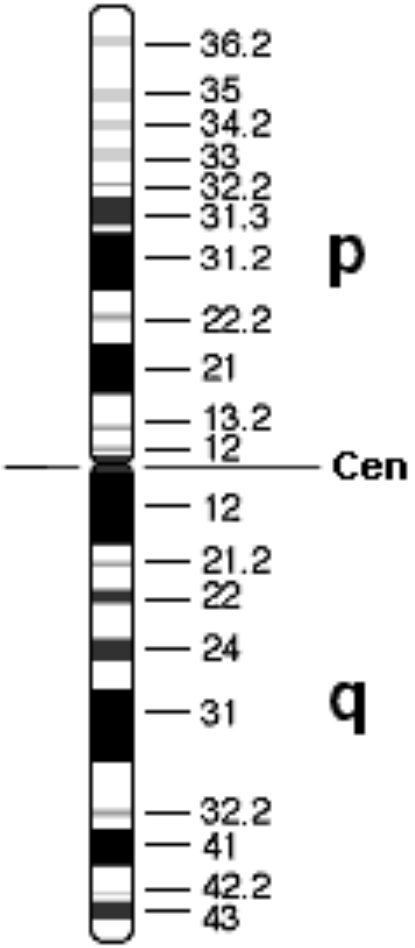


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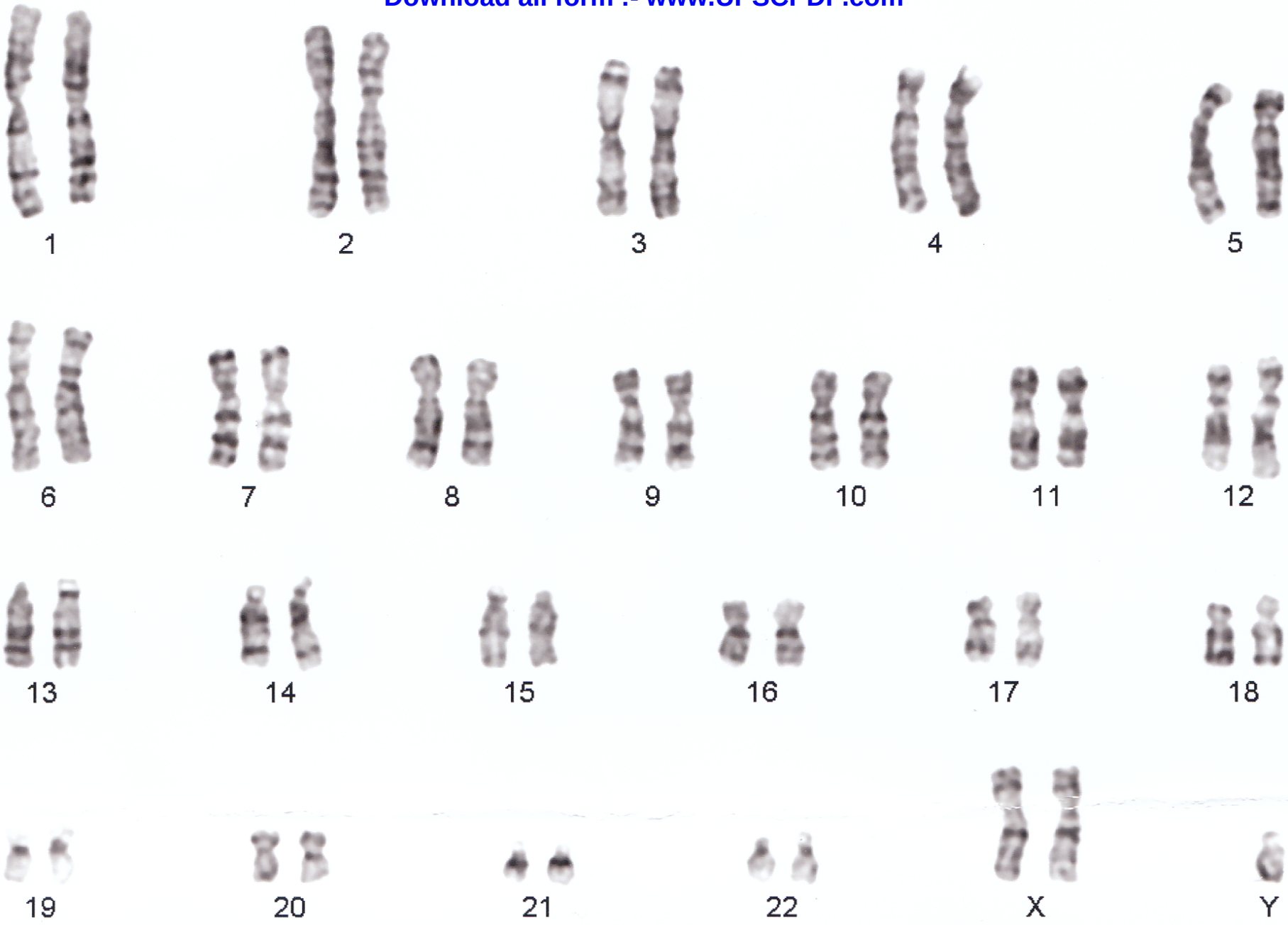
Ideogram

- The ideogram is basically a "chromosome map" showing the relationship between the short and long arms, centromere (cen), and in the case of acrocentric chromosomes the stalks (st) and satellites (sa).
- The specific banding patterns are also illustrated.
- Each band is numbered to aid in describing rearrangements.



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Chromosomes

- Normal human somatic cells have 46 chromosomes: 22 pairs, or ***homologs,*** of autosomes (chromosomes 1-22) and two sex chromosomes - This is called the ***diploid*** number.
- Females carry two X chromosomes (46,XX) while males have an X and a Y (46,XY).
- Germ cells (egg and sperm) have 23 chromosomes: one copy of each autosome plus a single sex chromosome - This is referred to as the ***haploid*** number.
- One chromosome from each autosomal pair plus one sex chromosome is inherited from each parent.
- Mothers can contribute only an X chromosome to their children while fathers can contribute either an X or a Y.

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